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Disparities in Stage at Presentation for Disaggregated Asian American, Native Hawaiian, and Pacific Islander Patients with Breast Cancer

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ABSTRACT

Introduction. Cancer is the leading cause of death among Asian American, Native Hawaiian, and Pacific Islander (AA&NHPI) populations, who are among the most diverse and rapidly growing racial/ethnic groups in the United States. Despite this population's diverse socioeconomic, cultural, and immigration backgrounds, little is known about the within-group disparities in breast cancer stages at diagnosis. This study aims to identify AA&NHPI subgroups at heightened risk for advanced breast cancer by analyzing detailed incidence rates and diagnostic stages across these diverse subgroups.

Methods. Patient data spanning from 2004 to 2020 were extracted from the National Cancer Database, focusing on individuals newly diagnosed with breast cancer. Individuals were disaggregated into specific AA&NHPI subgroups by self-reported country of origin. Ordinal logistic regression,

adjusting for clinical and sociodemographic factors, was used to calculate adjusted odds ratios (AORs). Higher adjusted odds ratios signify more advanced stage diagnoses.

Results. Of 3,001,173 patients with breast cancer, 99,128 (3.30%) identified as AA&NHPI. Compared with NHW patients, AA&NHPI patients were younger (median age, NHW: 62 years versus AA&NHPI: 57 years, $p < 0.001$) and more likely to be uninsured (NHW: 1.50% vs. AA&NHPI: 2.60%, $p < 0.001$). The largest AA&NHPI subgroups were Filipino ($n = 21,666$, 21.9% of AA&NHPI), Chinese ($n = 21,303$, 21.5%), Indian/Pakistani ($n = 20,295$, 20.5%), Vietnamese ($n = 6680$, 6.74%), and Korean ($n = 7776$, 7.84%). In aggregate, AA&NHPI patients were less likely to present with advanced stage at diagnosis relative to NHW patients (AOR 0.89, 95% confidence interval [CI] 0.88–0.90, $p < 0.001$). However, AA&NHPI disaggregation demonstrated that Hawaiian (AOR 1.15, 95% CI 1.09–1.22, $p < 0.001$), Hmong (AOR 1.70, 95% CI 1.28–2.27, $p < 0.001$), Laotian (AOR 1.37, 95% CI 1.17–1.61, $p < 0.001$), and Pacific Islander patients (AOR 1.24, 95% CI 1.17–1.32, $p < 0.001$) were more likely to present at a more advanced stage relative to NHW patients. Relative to Chinese Americans (the reference AA&NHPI group), all AA&NHPI ethnic groups had higher odds of advanced-stage diagnosis with the highest odds in Hmong (AOR 2.41, 95% CI 1.81–3.20, $p < 0.001$), Laotian (AOR 1.89, 95% CI 1.61–2.20, $p < 0.001$), Pacific Islander (AOR 1.71, 95% CI 1.60–1.82, $p < 0.001$), and Hawaiian (AOR 1.57, 95% CI 1.48–1.67, $p < 0.001$) patients.

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Conclusions. Our study characterizes disparities in the stage of breast cancer at presentation across disaggregated AA&NHPI groups, with notably advanced diagnoses in patients of Hmong, Laotian, Pacific Islander, Hawaiian, and Kampuchean descent. These findings underscore the need for disaggregated research exploring both risk factors and barriers to early diagnosis, as well as culturally tailored interventions that involve community leaders and leverage cultural norms to improve cancer screening and diagnosis.

Cancer remains the leading cause of death for Asian American and Pacific Islander (AA&NHPI) populations in the United States, accounting for more than 25% of all deaths.¹ The AA&NHPI population, encompassing 22 million individuals and 7% of the U.S. population, is one of the most diverse and fastest-growing groups in the nation.² This heterogeneity spans ethnic, cultural, historical, and socioeconomic dimensions, with roots spanning the Indian subcontinent and East and Southeast Asia, each with distinct immigration histories and patterns.² Notably, the greatest wealth gap of any group in the United States lies within the AA&NHPI community, with a median household income of \$119,000 among Indian Americans compared with \$44,000 among Burmese Americans.³ Disparities are evident in educational attainment as well, with 9% of Bhutanese Americans having a college degree, in contrast to 56% of Korean Americans.⁴ This great heterogeneity in social determinants of health may hold profound implications for the incidence, detection, and management of cancer within AA&NHPI subgroups.

AA&NHPI patients with breast cancer exhibit distinct patterns of disease compared with their non-Hispanic White counterparts owing to social determinants of health as well as unique genetic, environmental, and lifestyle factors.⁵ Young Chinese, Japanese, and Korean Asian Americans exhibit higher risks of hormone receptor-positive breast cancer compared with non-Hispanic White Americans, for example.⁶ Despite significant progress in characterizing breast cancer patterns in the AA&NHPI population, research has largely treated them as a monolithic group.

Recent studies have underscored the importance of disaggregating data to understand disparities in diseases, such as prostate, lung, colon, cervical, and skin cancers, among AA&NHPI populations.⁷⁻¹¹ This approach, which examines data by specific subgroups rather than broad racial categories, highlights the unique characteristics and needs of these diverse groups. Although increasingly recognized, historically, many studies have only analyzed data by race, which does not fully capture the diversity within the AA&NHPI communities. Moving toward a more detailed disaggregation can uncover important disparities in stage of disease at presentation.

METHODS

Data Source and Patients

The National Cancer Database (NCDB) is a nationwide, hospital-based cancer registry with a methodology that has been described previously. The extensive sample size of the NCDB allows for the analysis to be broken down by specific Asian American populations.^{12,13}

Our study evaluated a retrospective cohort analysis of female patients diagnosed with breast cancer between 2004 and 2020. The demographic scope of our study encompassed female patients self-identifying as non-Hispanic White, Thai, Asian Indian or Pakistani, Chinese, Vietnamese, Korean, Japanese, Filipino, Hawaiian, Laotian, Kampuchean, or Hmong. Of note, Indian and Pakistani were grouped together, as early years in the NCDB did not disaggregate these distinct subgroups. Because of the relatively small sample sizes of certain subgroups, we combined several Pacific Islander categories—including Micronesian, Chamorro, Guamanian, Polynesian, Tahitian, Samoan, Tongan, Melanesian, Fiji Islander, and New Guinean—into a single analytical cohort, while we recognize the broad diversity and cultural differences within this subgroup.¹⁴ Patients of Hawaiian descent were analyzed separately owing to a sufficiently large sample size within this subgroup.

Clinical and Sociodemographic Covariates

The outcome of interest was cancer stages at diagnosis, ranging from 0 (preinvasive/DCIS) to 4 (metastatic). We adjusted all models for clinical and sociodemographic covariates, including race/ethnicity, age, treatment facility type, median household income from the patient's ZIP code as a socioeconomic status indicator, the percentage of adults without a high school diploma in the patient's ZIP code as an educational attainment measure, Charlson–Deyo comorbidity score, insurance status, and diagnosis year. For income and education, we divided the data into quartiles according to the 2012 American Community Survey, corresponding to each patient's ZIP code.

Statistical Analysis

Pearson's chi-squared test and Kruskal-Wallis rank-sum test were used to compare categorical and continuous variables between different racial subgroups. Ordinal logistic regression was used to analyze the odds of patients presenting with increasingly advanced stages of breast cancer, adjusting for quantitative and categorical independent variables. We calculated two sets of adjusted odds ratios (AORs) and 95% confidence intervals (CIs). In the first model, we compared AA&NHPI racial subgroups to non-Hispanic

White patients to assess their likelihood of presenting with more advanced cancer stages. In the second model, focusing on the AA&NHPI subgroups, we used Chinese patients as the reference group, because they are the largest AA&NHPI subgroup in the United States. The entire population was categorized by receptor status—HR+ with HER2–, HER2+ regardless of estrogen/progesterone receptor status, and triple negative—and the analysis was then rerun for each group. We conducted these analyses by using Stata version 16.1 (StataCorp, College Station, TX) and R statistical software. The institutional review board exempted this study, because it used deidentified data.

RESULTS

Baseline Characteristics

Table 1 depicts clinical and sociodemographic characteristics of NHW patients and aggregated AA&NHPI subgroups. Of the 3,001,173 patients included in the study, 99,128 were in the aggregate AA&NHPI group. AA&NHPI patients were younger relative to NHW patients (median age, NHW: 62 years vs. AA&NHPI: 57 years, $p < 0.001$). AA&NHPI patients had higher rates of Medicaid (12% vs. 4.8%, $p < 0.001$), private insurance (60% vs. 52%, $p < 0.001$), and uninsured status (2.6% vs. 1.5%, $p < 0.001$) compared with NHW patients, whereas Medicare coverage was lower among AA&NHPI patients (23% vs. 40%, $p < 0.001$; Table 1). While NHW patients were most frequently treated at a Comprehensive Community Cancer Program (41%), AA&NHPI patients most often received treatment at an academic or research center (36%, $p < 0.001$). AA&NHPI patients were more represented in the highest zip-code level income bracket compared with their NHW counterparts (54% vs. 36%, $p < 0.001$). However, AA&NHPI patients were also more often residing in zip codes with low high school graduation rates (16% vs. 11% for NHW, $p < 0.001$). Additionally, AA&NHPI patients were slightly more likely to present with zero comorbidities (86% vs. 85%, $p < 0.001$). The largest AA&NHPI subgroups were Chinese ($n = 21,303$, 21.49% of AA&NHPI), Filipino ($n = 21,666$, 21.86%), Indian/Pakistani ($n = 20,295$, 20.47%), Vietnamese ($n = 6680$, 6.74%), and Korean ($n = 7776$, 7.84%). In aggregate, AA&NHPI patients presented more frequently with stage 0 disease (AA&NHPI: 24% vs. NHW: 19%, $p < 0.001$).

Table 2 presents a comparison of baseline characteristics of NHW and AA&NHPI groups, disaggregated into specific ethnic populations. Hmong and Laotian patients presented with stage 4 disease most frequently among all populations studied (9.3% and 6.9% respectively; Fig. 1). Hmong and Laotian patients were diagnosed at the youngest median age (52 and 53, respectively). Hmong (24%) and NHW (27%)

patients were least likely to be treated in an academic or research program ($p < 0.001$ for all comparisons).

Hmong (21%) and Laotian patients (28%) were least likely to live in the highest zip-code level income bracket, whereas Hmong (35%) and Kampuchean (35%) patients were most likely to live in zip codes with low rates of high school education. Hawaiian patients were the least likely to have no documented comorbidities (75%). Indian/Pakistani patients had the highest rate of uninsurance (4.5%), while Kampuchean patients had the highest rate of Medicaid (29%) ($p < 0.001$ for all comparisons).

Additionally, any hormone receptor status, HER2-positive was highest in Kampuchean (15%) and Laotian (13%) patients ($p < 0.001$). Hormone receptor-positive, HER2-negative was most prevalent among Hawaiian (34%), Japanese (30%), and NHW (30%) patients ($p < 0.001$). Triple-negative breast cancer rates were highest among Hmong (6.8%) and Laotian (7.5%) patients ($p < 0.001$).

Disparities in Stage at Presentation

Taken in aggregate, AA&NHPI patients were less likely to present with advanced stage at diagnosis (AOR 0.89, 95% CI 0.88–0.90, $p < 0.001$) relative to NHW patients (Supplementary Table 1). However, individual AA&NHPI populations displayed disparities in stage at presentation upon disaggregation.

In the primary analysis comparing AA&NHPI groups to non-Hispanic White patients (Table 3), Chinese (AOR 0.70, 95% CI 0.69–0.72, $p < 0.001$), Filipinos (AOR 0.90, 95% CI 0.88–0.93, $p < 0.001$), Japanese (AOR 0.78, 95% CI 0.76–0.81, $p < 0.001$), Koreans (AOR 0.93, 95% CI 0.90–0.97, $p = 0.001$), and Vietnamese (AOR 0.90, 95% CI 0.86–0.95, $p < 0.001$) were less likely to present at advanced stages compared with non-Hispanic White patients (Fig. 2). Conversely, Hawaiian (AOR 1.15, 95% CI 1.09–1.22, $p < 0.001$), Hmong (AOR 1.70, 95% CI 1.28–2.27, $p < 0.001$), Laotian (AOR 1.37, 95% CI 1.17–1.61, $p < 0.001$), and Pacific Islander (AOR 1.24, 95% CI 1.17–1.32, $p < 0.001$) patients were more likely to present at advanced stages compared with non-Hispanic White patients (Table 3).

In the subgroup analysis with Chinese patients as the referent, all analyzed subgroups were more likely to present at advanced stages compared with Chinese patients (Table 3; Fig. 3). The five ethnic groups with the highest adjusted odds ratios (AOR) were: Hmong (AOR 2.41, 95% CI 1.81–3.20, $p < 0.001$), Laotian (AOR 1.89, 95% CI 1.61–2.20, $p < 0.001$), Pacific Islander (AOR 1.71, 95% CI 1.60–1.82, $p < 0.001$), Hawaiian (AOR 1.57, 95% CI 1.48–1.67, $p < 0.001$), and Kampuchean (AOR 1.45, 95% CI 1.25–1.68, $p < 0.001$).

Among patients with available receptor status, the analysis was repeated with stratification by receptor status to reassess the AORs of advanced stage presentation for all AA&NHPI groups compared with NHW patients. Hawaiian, Hmong, Pacific Islander, and Laotian patients, who initially showed significantly higher odds of late-stage presentation compared with NHW patients, exhibited increased odds only when restricted to the HR-positive, HER2-negative disease cohort. Among these groups, only Pacific Islanders showed increased odds for the cohort restricted to HER2-positive disease (Tables 3 and 4). None of these groups experienced

significant and increased odds when restricted to the triple-negative cohort.

The receptor status stratification was continued for the staging analysis referent to Chinese patients. Filipino, Hmong, Laotian, Vietnamese, Hawaiian, Indian/Pakistani, Korean, Pacific Islander, and Thai patients, who initially showed significantly higher odds of late-stage presentation compared with Chinese patients, exhibited increased odds in the HR-positive, HER2-negative disease cohort. For the HER2-positive disease cohort, Filipino, Indian/Pakistani, and Pacific Islander patients showed significant

TABLE 1 Characteristics of baseline cohort stratified by aggregated AA&NHPI versus non-Hispanic White status, including stage at presentation, age, facility type, socioeconomic factors, comorbidity score, insurance status, and year of diagnosis (2004–2020), in the National Cancer Database

Characteristic	AA&NHPI <i>N</i> = 99,128 ^a	Non-Hispanic White <i>N</i> = 2,902,045 ^a	<i>p</i> ^b
Stage at presentation			
0	23,872 (24%)	562,911 (19%)	< 0.001
1	41,358 (42%)	1,362,263 (47%)	
2	22,907 (23%)	643,555 (22%)	
3	7544 (7.6%)	220,028 (7.6%)	
4	3447 (3.5%)	113,288 (3.9%)	
Age, years	57 (13)	62 (13)	< 0.001
Facility type			
Community cancer program	6548 (6.6%)	209,607 (7.2%)	< 0.001
Comprehensive community cancer program	31,458 (32%)	1,199,701 (41%)	
Academic or research program	36,054 (36%)	769,748 (27%)	
Integrated network cancer program	18,188 (18%)	606,112 (21%)	
Unknown	6880 (6.9%)	116,877 (4.0%)	
Zip-code level median household income			
< \$38,000	4570 (4.6%)	299,092 (10%)	< 0.001
\$38,000–\$47,999	10,169 (10%)	536,700 (18%)	
\$48,000–\$62,999	21,150 (21%)	709,199 (24%)	
\$63,000 +	53,725 (54%)	1,039,448 (36%)	
Unknown	9514 (9.6%)	317,606 (11%)	
Zip-code level percent without high school education			
21% or more	15,833 (16%)	313,979 (11%)	< 0.001
13%–20.9%	17,969 (18%)	565,897 (19%)	
7%–12.9%	28,443 (29%)	890,397 (31%)	
< 7%	27,380 (28%)	815,232 (28%)	
Unknown	9,503 (9.6%)	316,540 (11%)	
Charlson-Deyo comorbidity score			
0	85,465 (86%)	2,452,363 (85%)	< 0.001
1	10,877 (11%)	342,883 (12%)	
2	1811 (1.8%)	74,307 (2.6%)	
3 +	975 (1.0%)	32,492 (1.1%)	
Health insurance			
Uninsured	2601 (2.6%)	44,653 (1.5%)	< 0.001
Private insurance or managed care	59,476 (60%)	1,499,634 (52%)	
Medicaid	11,583 (12%)	137,986 (4.8%)	
Medicare	22,843 (23%)	1,149,118 (40%)	
Other government	965 (1.0%)	26,766 (0.9%)	
Unknown	1660 (1.7%)	43,888 (1.5%)	

Table 1 (continued)

Characteristic	AA&NHPI <i>N</i> = 99,128 ^a	Non-Hispanic White <i>N</i> = 2,902,045 ^a	<i>p</i> ^b
Year of diagnosis			
2004	3324 (3.4%)	130,519 (4.5%)	< 0.001
2005	3377 (3.4%)	134,166 (4.6%)	
2006	3504 (3.5%)	139,617 (4.8%)	
2007	3876 (3.9%)	145,715 (5.0%)	
2008	4261 (4.3%)	154,527 (5.3%)	
2009	4405 (4.4%)	163,229 (5.6%)	
2010	4988 (5.0%)	162,449 (5.6%)	
2011	5493 (5.5%)	170,249 (5.9%)	
2012	5814 (5.9%)	175,552 (6.0%)	
2013	6364 (6.4%)	183,266 (6.3%)	
2014	6756 (6.8%)	187,761 (6.5%)	
2015	7351 (7.4%)	192,234 (6.6%)	
2016	7592 (7.7%)	194,945 (6.7%)	
2017	8327 (8.4%)	200,029 (6.9%)	
2018	8182 (8.3%)	194,328 (6.7%)	
2019	8462 (8.5%)	200,647 (6.9%)	
2020	7052 (7.1%)	172,812 (6.0%)	
Subtype			
Any hormone receptor status, HER2-positive	7764 (7.8%)	170,902 (5.9%)	< 0.001
Hormone receptor-positive, HER2-negative	27,466 (28%)	858,839 (30%)	
Triple negative	3989 (4.0%)	123,764 (4.3%)	
Unknown	59,909 (60%)	1,748,540 (60%)	

^aData are numbers with percentages in parentheses or means with standard deviations in parentheses

^bPearson's chi-squared test; Wilcoxon rank-sum test

and increased odds of advanced diagnosis relative to Chinese patients. In the triple-negative disease cohort, Filipino, Japanese, Kampuchean, Korean, Vietnamese, Indian/Pakistani, and Pacific Islander patients showed significant and increased odds of advanced diagnosis relative to Chinese patients (Fig. 2; Table 5).

Additionally, receptor subtype analysis revealed significant results for ethnic groups that did not show significantly higher odds of late-stage presentation in the overall analysis relative to NHW patients. Filipinos (AOR 1.13, 95% CI 1.08–1.19, $p < 0.001$) and Indian/Pakistani patients (AOR 1.16, 95% CI 1.10–1.22, $p < 0.001$) had increased odds of advanced-stage diagnosis in the HR-positive, HER2-negative cohort. Filipinos (AOR 1.10, 95% CI 1.01–1.19, $p = 0.030$) and Indian/Pakistani patients (AOR 1.17, 95% CI 1.07–1.28, $p = 0.001$) also showed increased odds in the HER2-positive cohort. For the triple-negative cohort, Japanese (AOR 1.25, 95% CI 1.04–1.51, $p = 0.019$) and Kampuchean patients (AOR 2.07, 95% CI 1.08–3.96, $p = 0.028$) demonstrated significantly increased odds for advanced-stage diagnosis.

DISCUSSION

In this study of 3 million female patients with breast cancer, we found increased odds of advanced diagnosis for Hawaiian, Hmong, Laotian, and Pacific Islander populations in stage at presentation even after adjusting for a range of clinical and socioeconomic factors, including income, education, and comorbidity scores. Disaggregation revealed that all studied groups were more likely to present with a more advanced stage than Chinese patients, who served as the reference AA&NHPI group.

Our findings align with previous studies on disparities in breast cancer stages among aggregated AA&NHPI populations. Surveillance, Epidemiology, and End Results (SEER) studies found that Japanese patients were more likely to be diagnosed at earlier stages compared with non-Hispanic white patients, for example.^{15,16} Our work expands these findings to smaller groups, such as Hawaiian, Hmong, Laotian, and Pacific Islanders, who are often overlooked in research.¹⁷⁻²⁰

As demonstrated, AA&NHPI patients as an aggregated group present at a lower stage of breast cancer. However,

TABLE 2 Characteristics of baseline cohort stratified by disaggregated AA&NHP1 versus non-Hispanic White status, including stage at presentation, age, facility type, socioeconomic factors, comorbidity score, insurance status, and year of diagnosis (2004–2020), in the National Cancer Database

Characteristic	Non-Hispanic White N = 2,902,045 ^a	Chinese N = 21,303 ^a	Filipino N = 21,666 ^a	Hawaiian N = 4,438 ^a	Hmong N = 162 ^a	Indian/ Pakistani N = 20,295 ^a	Japanese N = 11,015 ^a	Kampuchean N = 639 ^a	Korean N = 7776 ^a	Laotian N = 536 ^a	Pacific Islander N = 3532 ^a	Thai N = 1086 ^a	Vietnamese N = 6680 ^a	p ^b
Stage at presentation														
0	562,911 (19%)	5941 (28%)	5528 (26%)	788 (18%)	23 (14%)	4418 (22%)	2684 (24%)	137 (21%)	1801 (23%)	91 (17%)	677 (19%)	247 (23%)	1537 (23%)	< 0.001
1	1,362,263 (47%)	9079 (43%)	8531 (39%)	1968 (44%)	51 (31%)	8296 (41%)	5250 (48%)	238 (37%)	3221 (41%)	196 (37%)	1370 (39%)	443 (41%)	2715 (41%)	
2	643,555 (22%)	4479 (21%)	5000 (23%)	1148 (26%)	49 (30%)	4997 (25%)	2194 (20%)	159 (25%)	1916 (25%)	150 (28%)	901 (26%)	255 (23%)	1659 (25%)	
3	220,028 (7.6%)	1255 (5.9%)	1744 (8.0%)	365 (8.2%)	24 (15%)	1791 (8.8%)	627 (5.7%)	70 (11%)	609 (7.8%)	62 (12%)	356 (10%)	86 (7.9%)	555 (8.3%)	
4	113,288 (3.9%)	549 (2.6%)	863 (4.0%)	169 (3.8%)	15 (9.3%)	793 (3.9%)	260 (2.4%)	35 (5.5%)	229 (2.9%)	37 (6.9%)	228 (6.5%)	55 (5.1%)	214 (3.2%)	
Age, years	62 (13)	57 (13)	59 (12)	59 (13)	52 (15)	55 (13)	62 (14)	54 (12)	55 (12)	53 (12)	57 (12)	55 (12)	55 (12)	< 0.001
Facility type														
Community cancer program	209,607 (7.2%)	798 (3.7%)	2011 (9.3%)	334 (7.5%)	6 (3.7%)	1066 (5.3%)	1346 (12%)	29 (4.5%)	326 (4.2%)	26 (4.9%)	221 (6.3%)	58 (5.3%)	327 (4.9%)	< 0.001
Comprehensive community cancer program	1,199,701 (41%)	6966 (33%)	7392 (34%)	1177 (27%)	56 (35%)	5972 (29%)	2914 (26%)	205 (32%)	2478 (32%)	176 (33%)	1110 (31%)	350 (32%)	2662 (40%)	
Academic or research program	769,748 (27%)	9008 (42%)	6811 (31%)	1536 (35%)	39 (24%)	8189 (40%)	3732 (34%)	234 (37%)	3042 (39%)	165 (31%)	1039 (29%)	364 (34%)	1895 (28%)	
Integrated network cancer program	606,112 (21%)	3143 (15%)	4388 (20%)	1120 (25%)	25 (15%)	3004 (15%)	2575 (23%)	93 (15%)	1354 (17%)	101 (19%)	895 (25%)	219 (20%)	1271 (19%)	
Unknown	116,877 (4.0%)	1388 (6.5%)	1064 (4.9%)	271 (6.1%)	36 (22%)	2064 (10%)	448 (4.1%)	78 (12%)	576 (7.4%)	68 (13%)	267 (7.6%)	95 (8.7%)	525 (7.9%)	
Zip-code level median household income														
< \$38,000	299,092 (10%)	1198 (5.6%)	739 (3.4%)	202 (4.6%)	30 (19%)	879 (4.3%)	224 (2.0%)	87 (14%)	293 (3.8%)	78 (15%)	253 (7.2%)	52 (4.8%)	535 (8.0%)	< 0.001
\$38,000–\$47,999	536,700 (18%)	2588 (12%)	2103 (9.7%)	391 (8.8%)	55 (34%)	1831 (9.0%)	920 (8.4%)	112 (18%)	623 (8.0%)	113 (21%)	455 (13%)	131 (12%)	847 (13%)	
\$48,000–\$62,999	709,199 (24%)	3771 (18%)	5015 (23%)	1183 (27%)	36 (22%)	3849 (19%)	2340 (21%)	175 (27%)	1620 (21%)	161 (30%)	924 (26%)	270 (25%)	1806 (27%)	
\$63,000 +	1,039,448 (36%)	12,462 (58%)	11,547 (53%)	1964 (44%)	34 (21%)	11,741 (58%)	6483 (59%)	196 (31%)	4531 (58%)	150 (28%)	1,473 (42%)	521 (48%)	2623 (39%)	
Unknown	317,606 (11%)	1284 (6.0%)	2262 (10%)	698 (16%)	7 (4.3%)	1995 (9.8%)	1048 (9.5%)	69 (11%)	709 (9.1%)	34 (6.3%)	427 (12%)	112 (10%)	869 (13%)	
Zip-code level percent without high school education														
≥ 21%	313,979 (11%)	4412 (21%)	3954 (18%)	291 (6.6%)	56 (35%)	2429 (12%)	985 (8.9%)	223 (35%)	794 (10%)	164 (31%)	599 (17%)	144 (13%)	1782 (27%)	< 0.001
13%–20.9%	565,897 (19%)	3628 (17%)	5220 (24%)	842 (19%)	34 (21%)	3100 (15%)	1394 (13%)	134 (21%)	1388 (18%)	138 (26%)	688 (19%)	177 (16%)	1226 (18%)	
7%–12.9%	890,397 (31%)	5688 (27%)	6334 (29%)	1476 (33%)	42 (26%)	5660 (28%)	3813 (35%)	135 (21%)	2178 (28%)	123 (23%)	1024 (29%)	328 (30%)	1642 (25%)	
< 7%	815,232 (28%)	6296 (30%)	3897 (18%)	1131 (25%)	23 (14%)	7116 (35%)	3775 (34%)	78 (12%)	2707 (35%)	77 (14%)	794 (22%)	325 (30%)	1161 (17%)	
Unknown	316,540 (11%)	1279 (6.0%)	2261 (10%)	698 (16%)	7 (4.3%)	1990 (9.8%)	1048 (9.5%)	69 (11%)	709 (9.1%)	34 (6.3%)	427 (12%)	112 (10%)	869 (13%)	

Table 2 (continued)

Characteristic	Non-Hispanic White N = 2,902,045 ^a	Chinese N = 21,303 ^a	Filipino N = 21,666 ^a	Hawaiian N = 4,438 ^a	Hmong N = 162 ^a	Indian/ Pakistani N = 20,295 ^a	Japanese N = 11,015 ^a	Kampuchean N = 639 ^a	Korean N = 7776 ^a	Laotian N = 536 ^a	Pacific Islander N = 3532 ^a	Thai N = 1086 ^a	Vietnamese N = 6680 ^a	p ^b
Charlson-Devo comorbidity score														
0	2,452,363 (85%)	19,374 (91%)	17,951 (83%)	3335 (75%)	136 (84%)	17,385 (86%)	9421 (86%)	557 (87%)	7037 (90%)	458 (85%)	2854 (81%)	982 (90%)	5975 (89%)	< 0.001
1	342,883 (12%)	1611 (7.6%)	2892 (13%)	734 (17%)	19 (12%)	2473 (12%)	1248 (11%)	62 (9.7%)	610 (7.8%)	61 (11%)	495 (14%)	91 (8.4%)	581 (8.7%)	
2	74,307 (2.6%)	218 (1.0%)	532 (2.5%)	229 (5.2%)	4 (2.5%)	305 (1.5%)	227 (2.1%)	13 (2.0%)	89 (1.1%)	12 (2.2%)	100 (2.8%)	7 (0.6%)	75 (1.1%)	
3 +	32,492 (1.1%)	100 (0.5%)	291 (1.3%)	140 (3.2%)	3 (1.9%)	132 (0.7%)	119 (1.1%)	7 (1.1%)	40 (0.5%)	5 (0.9%)	83 (2.3%)	6 (0.6%)	49 (0.7%)	
Health insurance														
Uninsured	44,653 (1.5%)	344 (1.6%)	445 (2.1%)	37 (0.8%)	4 (2.5%)	911 (4.5%)	73 (0.7%)	17 (2.7%)	331 (4.3%)	22 (4.1%)	128 (3.6%)	48 (4.4%)	241 (3.6%)	< 0.001
Private insurance	1,499,634 (52%)	12,900 (61%)	13,587 (63%)	2514 (57%)	70 (43%)	12,597 (62%)	6497 (59%)	308 (48%)	4328 (56%)	308 (57%)	1938 (55%)	685 (63%)	3744 (56%)	
or man-aged care														
Medicaid	137,986 (4.8%)	3218 (15%)	1889 (8.7%)	402 (9.1%)	49 (30%)	2464 (12%)	226 (2.1%)	183 (29%)	1110 (14%)	106 (20%)	545 (15%)	120 (11%)	1271 (19%)	
Medicare	1,149,118 (40%)	4415 (21%)	5152 (24%)	1380 (31%)	36 (22%)	3658 (18%)	3985 (36%)	120 (19%)	1758 (23%)	85 (16%)	799 (23%)	183 (17%)	1272 (19%)	
Other government	26,766 (0.9%)	47 (0.2%)	344 (1.6%)	57 (1.3%)	1 (0.6%)	128 (0.6%)	97 (0.9%)	2 (0.3%)	146 (1.9%)	8 (1.5%)	88 (2.5%)	29 (2.7%)	18 (0.3%)	
Unknown	43,888 (1.5%)	379 (1.8%)	249 (1.1%)	48 (1.1%)	2 (1.2%)	537 (2.6%)	137 (1.2%)	9 (1.4%)	103 (1.3%)	7 (1.3%)	34 (1.0%)	21 (1.9%)	134 (2.0%)	
Subtype														
Any hormone receptor status, HER2-positive	170,902 (5.9%)	1610 (7.6%)	1885 (8.7%)	316 (7.1%)	25 (15%)	1615 (8.0%)	500 (4.5%)	74 (12%)	619 (8.0%)	67 (13%)	287 (8.1%)	81 (7.5%)	685 (10%)	< 0.001
Hormone receptor-positive, HER2-negative	858,839 (30%)	5623 (26%)	5895 (27%)	1522 (34%)	34 (21%)	5690 (28%)	3352 (30%)	162 (25%)	2076 (27%)	120 (22%)	951 (27%)	296 (27%)	1745 (26%)	
Triple negative	123,764 (4.3%)	794 (3.7%)	676 (3.1%)	143 (3.2%)	11 (6.8%)	1081 (5.3%)	373 (3.4%)	31 (4.9%)	392 (5.0%)	40 (7.5%)	128 (3.6%)	49 (4.5%)	271 (4.1%)	
Unknown	1,748,540 (60%)	13,276 (62%)	13,210 (61%)	2457 (55%)	92 (57%)	11,909 (59%)	6790 (62%)	372 (58%)	4689 (60%)	309 (58%)	2166 (61%)	660 (61%)	3979 (60%)	

^aData are numbers with percentages in parentheses or means with standard deviations in parentheses^bPearson's chi-squared test; Kruskal-Wallis rank-sum test

FIG. 1 Unadjusted proportions of female patients with Stage 0-IV breast cancer at presentation, stratified by race or ethnicity

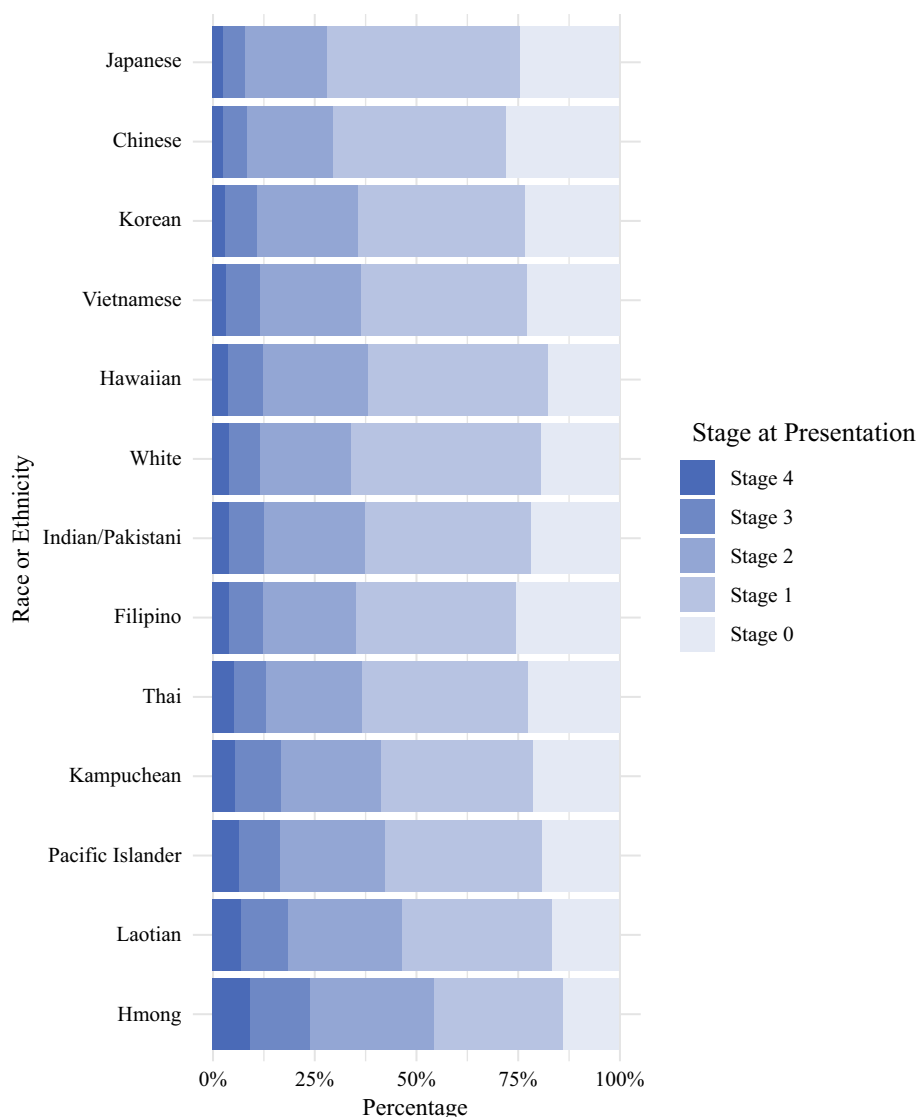
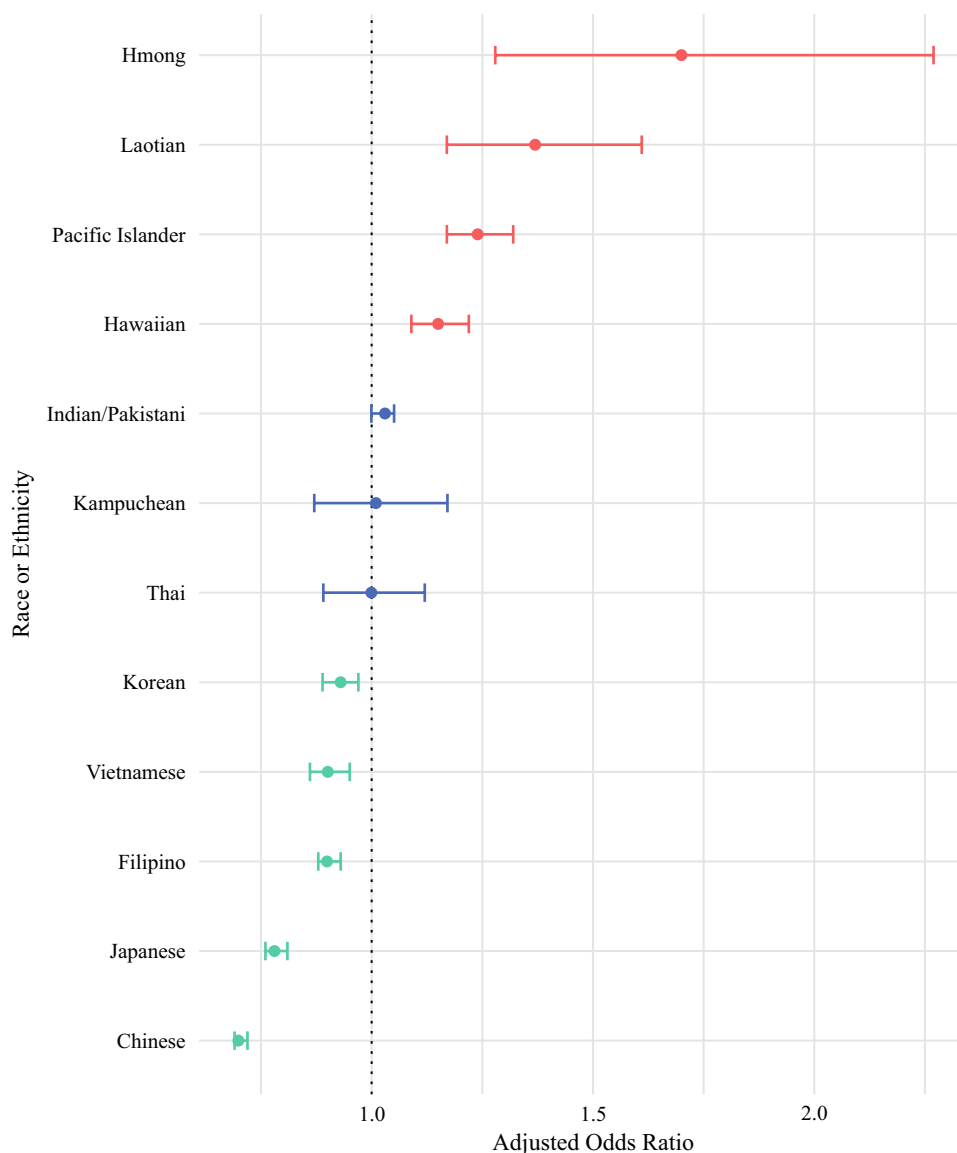


TABLE 3 Associations (AOR) and 95% CI (based on ordinal logistic regression) of race/ethnicity (relative to NHW and Chinese) and other selected patient and zip-code level characteristics with odds of presenting at progressively higher stage among AA&NHPI patients with breast cancer, 2004–2020, in the National Cancer Database

	Stage at presentation AOR (95% CI)	<i>p</i>	Stage at presentation AOR (95% CI)	<i>p</i>
Characteristic				
Non-Hispanic White	Ref.	.	Ref.	.
Chinese	0.70 (0.69–0.72)	< 0.001	Ref.	.
Filipino	0.90 (0.88–0.93)	< 0.001	1.26 (1.22–1.31)	< 0.001
Hawaiian	1.15 (1.09–1.22)	< 0.001	1.57 (1.48–1.67)	< 0.001
Hmong	1.70 (1.28–2.27)	< 0.001	2.41 (1.81–3.20)	< 0.001
Indian/Pakistani	1.03 (1.00–1.05)	0.051	1.40 (1.35–1.45)	< 0.001
Japanese	0.78 (0.76–0.81)	< 0.001	1.09 (1.05–1.14)	< 0.001
Kampuchean	1.01 (0.87–1.17)	0.865	1.45 (1.25–1.68)	< 0.001
Korean	0.93 (0.90–0.97)	0.001	1.27 (1.21–1.33)	< 0.001
Laotian	1.37 (1.17–1.61)	< 0.001	1.89 (1.61–2.20)	< 0.001
Pacific Islander	1.24 (1.17–1.32)	< 0.001	1.71 (1.60–1.82)	< 0.001
Thai	1.00 (0.89–1.12)	0.999	1.36 (1.22–1.52)	< 0.001
Vietnamese	0.90 (0.86–0.95)	< 0.001	1.27 (1.21–1.34)	< 0.001

AOR adjusted odds ratio, CI confidence interval

FIG. 2 Associations (aORs) and 95% confidence intervals (based on ordinal logistic regression) of race/ethnicity (relative to NHW patients) and other selected patient and zip-code level characteristics with odds of presenting at progressively higher stage among AANHPI patients with breast cancer, 2004–2020, in the National Cancer Database



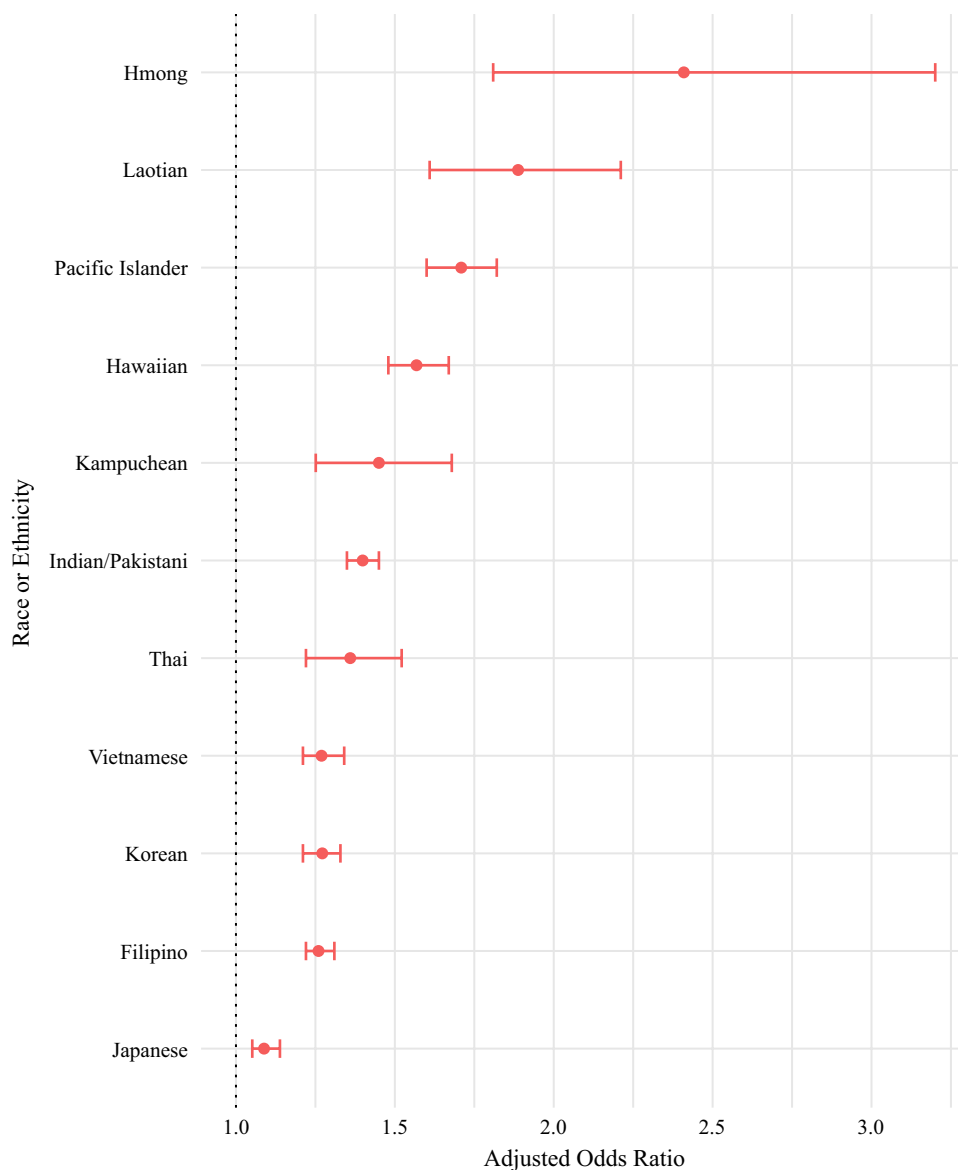
disaggregating the data reveals disparities among various AA&NHPI populations. We identified later stages at presentation for Hawaiian, Hmong, Laotian, and Pacific Islander patients compared with NHW patients, with Hmong patients nearly twice as likely to have advanced diagnoses. These disparities were previously undocumented owing to small sample sizes and merging into broader AA&NHPI subgroups.^{21,22} This study further identified a compounded disparity in breast cancer outcomes for Hmong and Laotian patients, who exhibit both the highest prevalence of triple-negative breast cancer (TNBC)—a subtype associated with poor prognosis—and the highest rates of Stage 4 disease at presentation. These findings are magnified by the observation that Hmong and Laotian patients are diagnosed at a median age nearly 10 years younger than NHW patients. These results emphasize the need for further investigation into genetic predispositions, environmental exposures, and

systemic barriers to early detection, along with the implementation of culturally tailored interventions to address these disparities. One-size-fits-all screening programs may not address the specific needs of smaller, heterogeneous communities and their access barriers.

Attitudes toward mammogram screening may play a role in presentation rates among Asian-American patients compared with NHW patients. Contributing factors may include health literacy, financial and insurance limitations, cultural and personal beliefs, fear and embarrassment, self-efficacy, religious and psychological aspects, time constraints, fatalism, healthcare professional advice, communication barriers, and social support. These factors contribute to lower mammography rates among AA&NHPI patients compared with NHW patients.²³⁻³¹

Disparities in breast cancer stage at diagnosis among Hawaiian, Hmong, Laotian, and Pacific Islander AA&NHPI

FIG. 3 Associations (aORs) and 95% confidence intervals (based on ordinal logistic regression) of race/ethnicity (relative to Chinese American patients) and other selected patient and zip-code level characteristics with odds of presenting at progressively higher stage among AANHPI patients with breast cancer, 2004–2020, in the National Cancer Database



subgroups may be influenced by specific cultural, linguistic, and socioeconomic barriers. Native Hawaiians encounter various healthcare barriers, including cultural challenges, such as limited access to cancer information and health-related anxiety; socioeconomic issues, such as poverty; systemic obstacles, such as lack of insurance and limited access to cancer care; and geographic difficulties owing to the centralization of cancer resources in Honolulu.³² In the Hmong community, social marginalization and refugee status contribute to mistrust of institutions, while cultural stigmas around women's health contribute to advanced diagnoses.³³ Many Laotian women serve as primary caregivers, with 40% in multigenerational households.³⁴ This role imposes time, financial, and logistical challenges, leading to treatment delays. Pacific Islander Americans, especially from

outer islands, such as the Marshall Islands, face geographic isolation from healthcare facilities, causing further delays.³⁵

Addressing disparities in breast cancer diagnosis among Hmong, Laotian, and Pacific Islander communities requires culturally informed interventions. For example, the “Life is Precious” program in the Hmong community incorporated family decision-making and shamans to increase women's intent to obtain mammograms and men's intent to support screening and treatment.³⁶ Future interventions for the Hmong community should collaborate further with religious leaders and involve male family members in the education and care-seeking process.

Another successful intervention is the PATH for Women program, which successfully increased breast cancer screening rates in the Laotian community.³⁷ The program addressed challenges, such as low literacy rates and limited

TABLE 4 Associations (AORs) and 95% CI (based on ordinal logistic regression) of race/ethnicity (relative to NHW) and other selected patient and zip-code level characteristics with odds of presenting at progressively higher stage among AA&NHPI patients with breast

cancer, 2004–2020, in the National Cancer Database, stratified by hormone and HER2 receptor status: (a) hormone receptor-positive, HER2-negative; (b) any hormone receptor status, HER2-positive; and (c) triple negative

Characteristic	Hormone receptor-positive, HER2-negative	<i>p</i>	Any hormone receptor status, HER2-positive	<i>p</i>	Triple negative	<i>p</i>
	AOR (95% CI)		AOR (95% CI)		AOR (95% CI)	
Non-Hispanic White	Ref.	.	Ref.	.	Ref.	.
Chinese	0.84 (0.80–0.88)	< 0.001	0.83 (0.76–0.91)	< 0.001	0.79 (0.70–0.91)	0.001
Filipino	1.13 (1.08–1.19)	< 0.001	1.10 (1.01–1.19)	0.030	1.13 (0.98–1.30)	0.083
Hawaiian	1.29 (1.17–1.42)	< 0.001	0.97 (0.80–1.19)	0.776	0.99 (0.74–1.34)	0.964
Hmong	3.13 (1.72–5.71)	< 0.001	1.60 (0.78–3.25)	0.197	0.94 (0.32–2.69)	0.900
Indian/Pakistani	1.16 (1.10–1.22)	< 0.001	1.17 (1.07–1.28)	0.001	1.00 (0.89–1.11)	0.934
Japanese	0.81 (0.75–0.86)	< 0.001	0.93 (0.79–1.09)	0.356	1.25 (1.04–1.51)	0.019
Kampuchean	0.79 (0.58–1.06)	0.118	1.16 (0.77–1.74)	0.488	2.07 (1.08–3.96)	0.028
Korean	1.05 (0.97–1.14)	0.233	0.98 (0.85–1.14)	0.827	1.09 (0.91–1.31)	0.344
Laotian	1.78 (1.28–2.47)	0.001	0.83 (0.53–1.30)	0.418	1.37 (0.76–2.48)	0.298
Pacific Islander	1.41 (1.25–1.58)	< 0.001	1.59 (1.29–1.96)	<0.001	1.16 (0.84–1.60)	0.378
Thai	1.22 (0.98–1.52)	0.072	1.06 (0.72–1.54)	0.772	1.21 (0.74–2.00)	0.451
Vietnamese	1.04 (0.95–1.14)	0.392	0.95 (0.83–1.09)	0.477	1.05 (0.84–1.31)	0.668

AOR adjusted odds ratio, CI confidence interval

TABLE 5 Associations (AOR) and 95% CI based on ordinal logistic regression) of race/ethnicity (relative to Chinese) and other selected patient and zip-code level characteristics with odds of presenting at progressively higher stage among AA&NHPI patients with breast

cancer, 2004–2020, in the National Cancer Database, stratified by hormone and HER2 receptor status: (a) hormone receptor-positive, HER2-negative; (b) any hormone receptor status, HER2-positive; and (c) triple negative

Characteristic	Hormone receptor-positive, HER2-negative	<i>p</i>	Any hormone receptor status, HER2-positive	<i>p</i>	Triple negative	<i>p</i>
	AOR (95% CI)		AOR (95% CI)		AOR (95% CI)	
Chinese	Ref.	.	Ref.	.	Ref.	.
Filipino	1.33 (1.24–1.43)	< 0.001	1.30 (1.15–1.48)	< 0.001	1.43 (1.18–1.74)	< 0.001
Hawaiian	1.51 (1.35–1.69)	< 0.001	1.16 (0.93–1.45)	0.194	1.21 (0.87–1.69)	0.268
Hmong	3.82 (2.07–7.06)	< 0.001	2.10 (1.01–4.36)	0.047	1.26 (0.43–3.69)	0.674
Indian/Pakistani	1.36 (1.26–1.46)	< 0.001	1.40 (1.23–1.59)	<0.001	1.25 (1.05–1.49)	0.014
Japanese	0.94 (0.86–1.03)	0.167	1.05 (0.87–1.27)	0.607	1.53 (1.20–1.94)	0.001
Kampuchean	0.95 (0.70–1.30)	0.764	1.51 (0.99–2.31)	0.057	2.76 (1.40–5.45)	0.003
Korean	1.22 (1.10–1.34)	< 0.001	1.19 (1.00–1.41)	0.052	1.35 (1.08–1.70)	0.009
Laotian	2.19 (1.56–3.08)	< 0.001	1.03 (0.65–1.63)	0.911	1.69 (0.91–3.14)	0.096
Pacific Islander	1.67 (1.47–1.91)	< 0.001	1.97 (1.56–2.48)	<0.001	1.43 (1.00–2.04)	0.048
Thai	1.44 (1.15–1.80)	0.002	1.27 (0.86–1.88)	0.239	1.45 (0.85–2.45)	0.173
Vietnamese	1.24 (1.12–1.38)	< 0.001	1.17 (1.00–1.38)	0.057	1.35 (1.04–1.76)	0.024

AOR adjusted odds ratio, CI confidence interval

Laotian media channels by emphasizing door-to-door and face-to-face outreach using visual-based educational materials. Future interventions could replicate this approach, prioritizing personal connections and visual aids. Additionally, deploying mobile health units with skilled staff and educational materials can enhance outreach. To establish

community-based networks, interventions should engage local leaders, partner with community organizations, and host regular health education events. Creating media channels could involve collaborating with local radio and TV stations to produce content in Laotian and using social media platforms to share educational information effectively.

Insights from a focus group study with 66 Samoan church members highlight the effectiveness of culturally aligned, church-based cancer awareness programs.³⁸ The study found strong support within the Samoan community for training pastors to frame cancer not as a spiritual illness but as a physical ailment that can be treated. This approach shifts the perception of cancer away from being seen as an incurable spiritual weakness. Future interventions for Pacific Islander communities can learn from this model by involving spiritual and community leaders in health messaging. Leveraging the community's hierarchical structure and deep spiritual beliefs, collaborations with religious institutions and existing community structures can enhance the acceptance and impact of health programs.

Finally, focus groups with Native Hawaiian cancer survivors highlighted the importance of Native Hawaiian traditions, such as support from friends, family, community members, and spiritual beliefs, in coping with cancer diagnosis and treatment.³⁹ The success of the PILI 'Ohana Project (POP), a community-based participatory research initiative addressing obesity and diabetes disparities, demonstrates the effectiveness of culturally adapted health programs. POP utilized images of familiar cultural foods, recruited community peer educators, and emphasized family and community involvement.⁴⁰

Community-level initiatives, such as cost-free screening services, mobile clinics, and educational programs, can enhance access for those facing financial or logistical challenges. For example, the Hawai'i Breast and Cervical Cancer Control Program (BCCCP) significantly increased screening rates among Micronesian women from 18% to 90% through tailored educational materials and personalized outreach.⁴¹ Clinically, it is essential to prioritize cultural competency training for healthcare professionals and enhance diversity in physician training to better serve AA&NHPI patients.⁴² Additionally, hospitals should train more interpreters to ensure patients and families can effectively communicate their beliefs and values.⁴³ On a federal level, interventions should include compensation for lost wages and costs incurred for screenings. Furthermore, investing in broadband infrastructure and programs like the Affordable Connectivity Program is crucial to improving telehealth access.⁴⁴

The study's limitations include the aggregation of several ethnicities under the Pacific Islander umbrella term, as well as the grouping together of Indian/Pakistani populations, which may mask critical cultural and socioeconomic differences within these diverse subgroups. Second, this study cannot establish causality with regards to the drivers of disparities in breast cancer stage at diagnosis owing to its retrospective nature. Third, nativity status may be associated with cancer outcomes, as has been shown for many immigrant groups in California but is unfortunately not available in the NCDB.²³ Lastly, while our study accounts for several

covariates, including income, education, and comorbidity scores, full clinical details and important qualitative factors including illness experiences and caregiving responsibilities were not available; thus, our findings may not encompass the full spectrum of factors involved in within-group differences.

In conclusion, our study provides important insights with regards to the complex relationship between the country of origin and breast cancer stage at presentation among AA&NHPI subgroups. The identified disparities emphasize the urgent need for targeted and culturally tailored interventions to ensure equitable access to timely breast cancer screening, detection, and treatment. Finally, we draw attention to the need for sustained research on the diverse cultural, environmental, biological, and socioeconomic forces driving within-group disparities. In combination with the delivery of comprehensive, holistic interventions, this approach may alleviate the high burden of breast cancer among marginalized AA&NHPI communities.

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